SAHIL JHAWAR

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EDUCATION

Universität Potsdam *October 2021 - June 2025*
Master of Science in Astrophysics *Potsdam, Germany*
Grade*: 1.7 (gut/good)
Thesis

Christ (Deemed to be University) *July 2018 - June 2021*
Bachelor of Science (Physics, Maths and Electronics) *Bengaluru, India*
GPA: 3.68/4.0

Final Year Project: Gesture Controlled System based on Raspberry Pi Pico *Report*

SKILLS

Technical Languages	Python, C\C++, LaTeX, MATLAB, SQL
Technical Skills	Git (-Lab and -Hub), Unix based OS and Windows, Bayesian Inference, Data analysis and visualisation, DevOps, Docker, Apptainer/Singularity
Packages and tools	NumPy, SciPy, Pandas, Matplotlib, Astropy, Bilby, Seaborn, Sphinx, TensorFlow, scikit-learn, JAX, PyTorch, CuPy, Streamlit, Selenium, bs4, PyBind, CI\CD, Parallel computing

WORK EXPERIENCE

GFZ Helmholtz Centre for Geosciences *July 2024 - Present*
Research Scientist - February 2026 - Present *Potsdam, Germany*
Research Assistant - July 2025 - December 2025
Student Assistant - July 2024 - June 2025
Software Developer in the Section 1.5 - Space Physics and Space Weather

- Developed a real-time ETL pipeline and testing framework for processing NASA/JAXA spacecraft data, implementing over 5,000 lines of code.
- Contributed to the development of EL-PASO, an open source Python framework for processing and standardizing multi-mission satellite particle observations (paper under review).
- Implemented CUDA/CuPy acceleration for interpolation routines, achieving more than a 10× speedup compared to NumPy/SciPy implementations.
- Built reproducible Apptainer/Singularity workflows for deployment on the European Space Agency's HPC infrastructure.
- Developed software deliverables for the European Space Agency as part of funded research projects.
- Developed orchestration and utility codes, reducing maintenance overhead by approximately 50%, and implemented structured logging, monitoring, and automated validation to improve debugging, reliability, and fault detection in operational space weather forecasting pipelines.
- Maintaining the section's operational Kp forecasting pipeline and developed the Kp alert system.

Leibniz-Institut für Astrophysik Potsdam (AIP) *December 2022 - March 2024*
Student Assistant *Potsdam, Germany*
Student Assistant in the Project Management team for building (integration, installation and maintenance) the 4-metre Multi-Object Spectroscopic Telescope *4MOST*.

*See here on how to interpret German grade: Welche Form der Benotung wird an der Uni Potsdam verwendet?

- Implemented *4MOST* commissioning procedures during 3 week coding campaign (November 2023)
- Utilized technical skills in troubleshooting and problem-solving within the specialized environment of an astronomical integration hall.
- Proactively managed procurement activities, liaising with vendors to procure essential large-scale items for the integration hall.

RESEARCH EXPERIENCE

Theoretical Astrophysics Group, Universität Potsdam *March 2023 - June 2025*
Master's Thesis *Potsdam, Germany*

Master's thesis student under **Prof. Dr. Tim Dietrich**.

- Maintaining and contributing to software infrastructure with over 60 forks, 40+ stars, and 15,000+ Conda downloads.
- Developed Bayesian models for analyzing astronomical datasets using NMMA, focusing on optimizing parameter estimation.
- Quantified and minimized non-stationary uncertainties by 2-10x in astrophysical parameter estimation and predictions.
- Designed a SHA-256 hash-based system to verify and automate file replacement for data consistency required for astrophysical simulations.
- Presented my findings as the first author on one research paper and as a co-author on another.

Universität Potsdam & Astronomical Institute of CAS *August 2022 - September 2022*
Participant *Prague, Czech Republic*

- Utilized the Perek 2m telescope at the Ondřejov Observatory to acquire stellar spectra
- Performed data reduction and analysis using IRAF Python for data visualization

IUCAA & NCRA-TIFR *December 2020 - January 2021*
Winter Student *Pune, India*

Radio Astronomy Winter School 2020: Learned basics of radio astronomy, from theory to instrumentation and computational aspects. Workshop enabled me to work on few physical experiments and on GMRT and ORT data of Vela pulsar.
Detailed reports can be found here

Krittika, IIT Bombay *May 2020 - August 2020*
Summer Project Intern *Mumbai, India*

Developed a Python package to analyse various types of binary star system.

PUBLICATIONS

1. **Sahil Jhawar**, Thibeau Wouters, Peter T. H. Pang, et al. "Data-driven approach for modeling the temporal and spectral evolution of kilonova systematic uncertainties". In: *Phys. Rev. D* 111 (4 2025), p. 043046. DOI: `10.1103/PhysRevD.111.043046`.
2. M M Desai, D Chatterjee, **S Jhawar**, et al. "Rapid parameter estimation for kilonovae using likelihood-free inference". In: *Monthly Notices of the Royal Astronomical Society* 541.3 (June 2025), pp. 2619–2630. DOI: `10.1093/mnras/staf1045`.
3. T. Hussenot-Desenonges, T. Wouters, N. Guessoum, et al. *Multi-band analyses of the bright GRB 230812B and the associated SN2023pel*. 2023. arXiv: `2310.14310 [astro-ph.HE]`.

TECHNICAL PROJECTS

SWVO
contributor and maintainer
Tools for downloading and reading space-weather data and geomagnetic indices.

EL-PASO
contributor and maintainer
EL-PASO is a Python framework designed to streamline the download, processing, and saving of satellite particle observation data.

reuseify
Automate REUSE license annotation from git history.

Kp Alert
Python-based system for monitoring forecasted Kp index and sending automated alerts.

NMMA
former contributor and maintainer
A Pythonic library for probing nuclear physics and cosmology with multimessenger analysis.

nbody
Direct N-body code with C++ and JAX backends.

Gstrain
Interactive Gravitational Wave strain viewer using Streamlit & Plotly. Check it out here

BibQuest
Automagically fill your bib file just using bib keys

uptime
A Github Action and Pages based uptime monitor in Python.

ACADEMIC VISITS AND TRAINING

- **2026** Summer school on Machine Learning for Space 2026 at KTH Stockholm, Sweden
- **2026** High-performance computing with Python at Forschungszentrum Jülich (online)
- **2024** Fully funded visiting fellowship by European Union's Horizon 2020 Programme under the AHEAD 2020 hosted by Dr. Mattia Bulla at University of Ferrara, Italy
- **2024** Fellowship of €500 by ESA for dotAstronomy 2024, ESAC Madrid, Spain

PRIZES, AWARDS AND HONORS

- **2023** Scholarship of ₹35,000 by Badrilal Soni Maheshwari Shiksha Sahyog Kendra, India
- **2021** Awarded co-curricular activities based scholarship from Christ (Deemed to be University)
- **2021** Won numerous inter and intra collegiate events, and successfully lead a team of players resulting in various rolling trophies for the department
- **2020** Gold honor in International Astronomy and Astrophysics Competition (IAAC), secured 38/40 and placed in the top 1% worldwide